

# A Morgan Stanley Banking Data Analysis

- Name :- Ms.Ratan Ghanghoriya
- Batch :- February 2026
- Course :- Data Science
- Video Link:-  
<https://www.loom.com/share/2cc33221464844ba950e955d2bcf28ad>

## Project Overview

This project analyzes JP Morgan banking transaction data using Python. The main goal of the project is to understand customer transaction behavior, identify financial risks, and provide useful business recommendations. The analysis includes data cleaning, transaction analysis, customer profiling, financial risk analysis, data visualization, and hypothesis testing. The findings help the bank understand customer behavior and support better business decisions.

### Objective

- Analyze banking data using Python
- Understand customer behavior.
- Identify financial risk.
- Study transaction trends.
- Suggest better business decisions.

### Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab

### Project Workflow

- Data Cleaning
- Transaction Analysis
- Customer Profile Building
- Financial Risk Identification
- Data Visualization
- Hypothesis Testing

### Key Findings & Insights

- The dataset contains **800 records and 15 columns**. It was clean, with **no missing values or duplicate records..**
- The average transaction amount is **55,195**, and the average account balance is **72,764**.

- **Savings Account** has the highest transaction amount and is the most frequently used banking product.
- The analysis identified **top-performing, low-performing, and dormant accounts**, helping the bank understand customer activity.
- Customers were grouped into **High, Medium, and Low activity** levels, and high-value customers were identified.
- The hypothesis test showed **P-value = 0.0801 (> 0.05)**, so there is **no statistically significant difference** between the compared customer groups

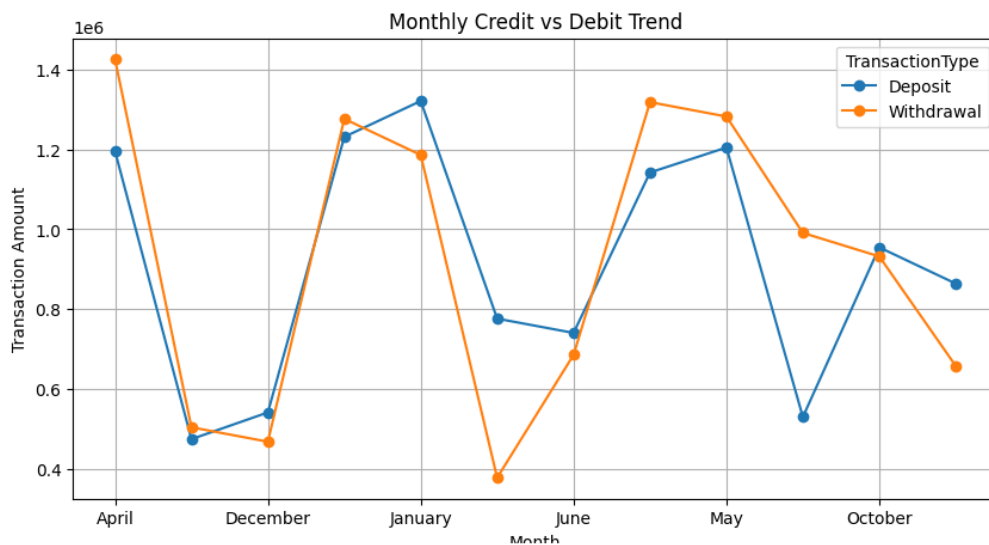
### What Drives Customer Transaction Behavior?

- Savings Account has the highest transaction amount.
- Customer transactions change from month to month.
- Deposits and withdrawals are different every month.
- High-activity customers make more transactions.
- Account type and transaction frequency affect customer behavior.

### What Drives Financial Risk?

- High-risk customer accounts were identified during the analysis.
- Some customer accounts showed **large withdrawal amounts**, which require monitoring.
- Inactive (dormant) customer accounts were identified and should be checked regularly.
- The correlation heatmap shows that the financial variables have **very weak relationships**, so financial risk cannot be explained by a single factor.
- The hypothesis test showed **P-value = 0.0801 (> 0.05)**, which means there is **no statistically significant difference** between the compared customer groups

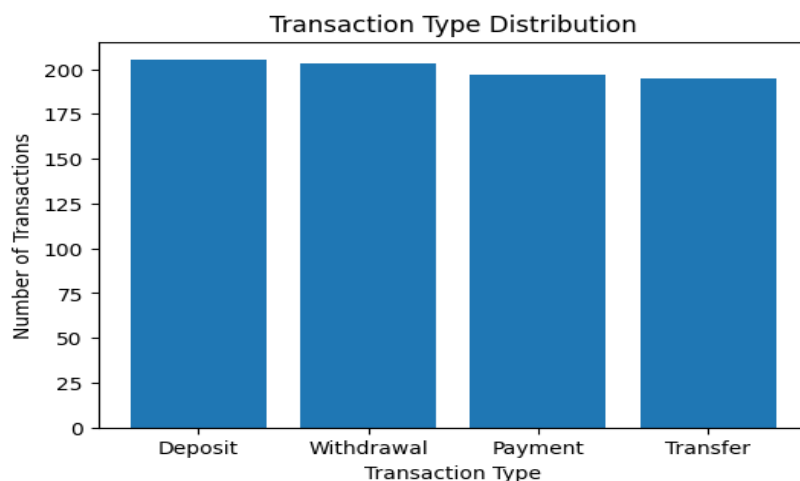
## Chart 1 :- Monthly Credit vs Debit Trend



### Summary

- This chart compares monthly **Credit** and **Debit** transactions.
- Credit transactions are higher in **February, July, September, October, and December**.
- Debit transactions are higher in **March, January, April, and May**.
- The chart shows that customer **saving and spending patterns change during different months**.
- The bank can use this information to monitor transaction trends and plan better customer services.

## Chart 2 :- Transaction Type Distribution

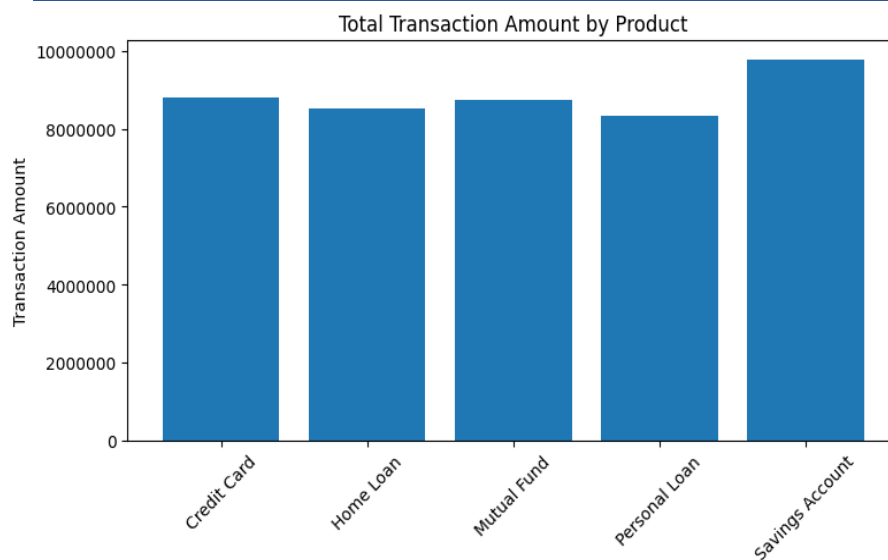


## Summary

- The chart shows the distribution of different transaction types.
- **Deposit, Withdrawal, Payment, and Transfer transactions are almost equally distributed** in the dataset.
- This shows that customers use different transaction types regularly.
- The bank should continue monitoring all transaction types to **improve customer service** and **detect unusual activities**

## Chart 3 : Total Transaction Amount by Product

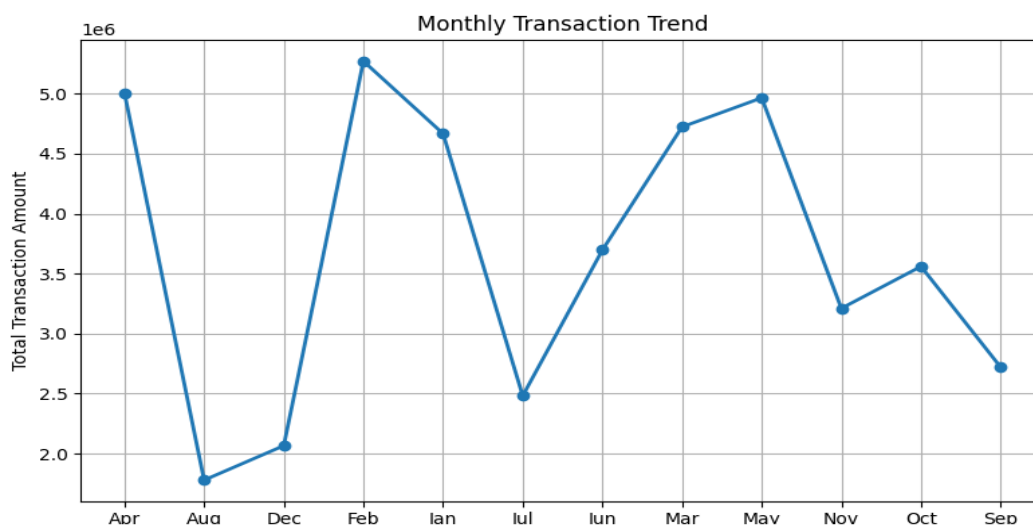
...



## Summary

- The chart compares the total transaction amount for different banking products.
- **Savings Account has the highest transaction amount.**
- **Personal Loan has the lowest transaction amount.**
- This shows that customers **use Savings Accounts more than other banking products.**
- The bank can promote lower-performing products to increase customer usage.

## Chart 4: Monthly Transaction Trend



### Summary

- The chart shows the monthly transaction trend.
- **February has the highest transaction amount, while August has the lowest.**
- Customer transaction activity changes from month to month.
- The bank can use this trend to improve planning and customer services.

### Data-backed Recommendations

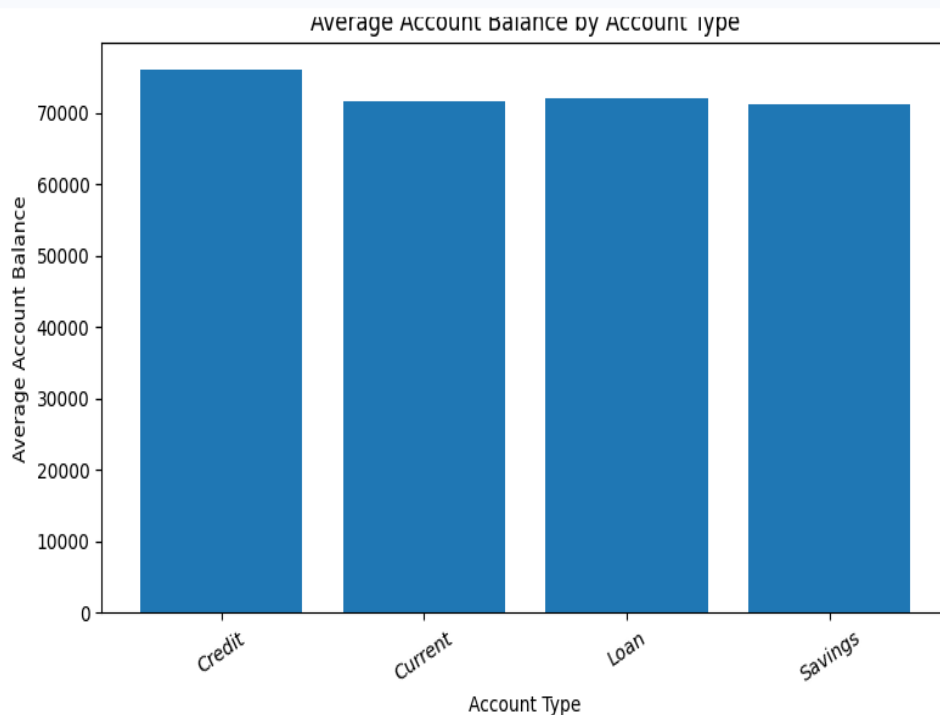
- Focus more on **Savings Account** because it has the highest transaction amount.
- Contact **dormant customers** and encourage them to start using their accounts again.
- Be ready for **high transaction months** like February by planning staff and banking services.

- Review **months with higher withdrawals** to understand why more money is going out.
- Give **special offers to customers with low account balances** to encourage saving.
- Check **high-risk accounts regularly** to reduce future financial risk.
- Use **customer activity and transaction history together** while making business decisions.

## Key Results

- The dataset was clean with **no missing values or duplicate records**.
- **Savings Account** recorded the highest total transaction amount.
- **February** had the highest transaction activity, while **August** had the lowest.
- Some accounts became **inactive** because they had no transactions for **60 days or more**.
- The heatmap showed **very weak relationships** between the financial variables.
- The hypothesis test found **no statistically significant difference** between high- and low-volume transaction accounts.

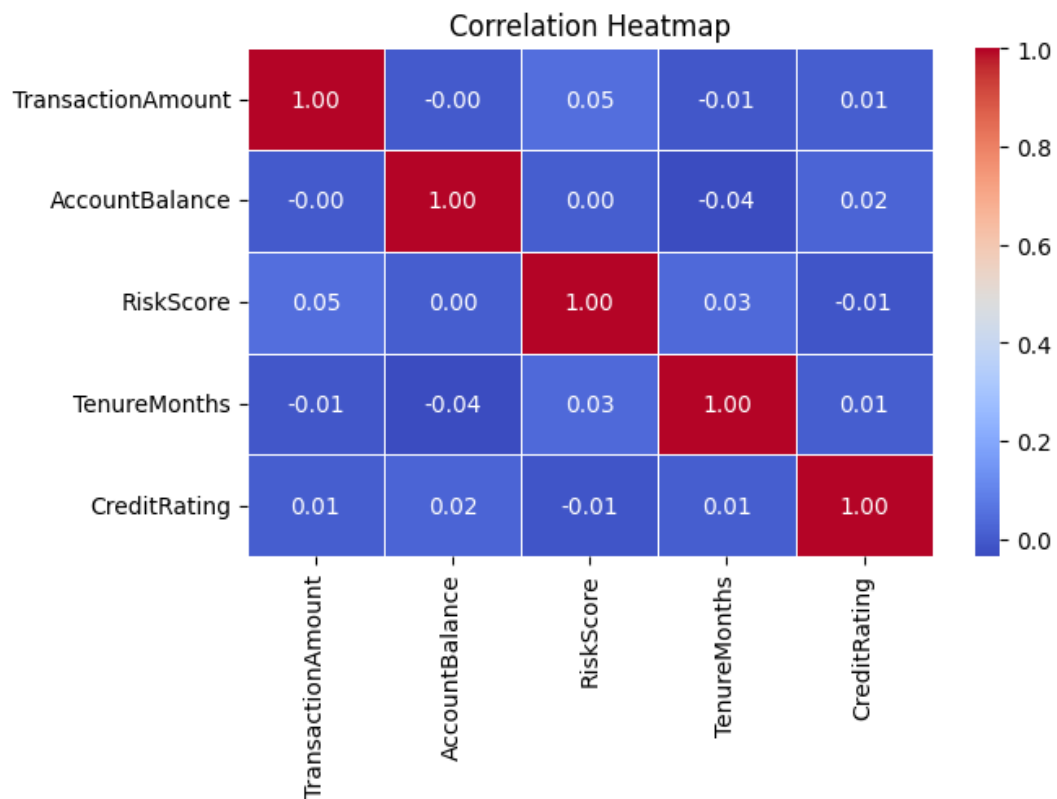
## Average Balance by Account Type



### Result:

**Credit accounts have the highest average balance, while the other account types have almost similar balances.**

## Correlation Heatmap



### **Result:**

**Most variables have very weak correlation, so customer behaviour cannot be explained by one factor alone.**

## Hypothesis Testing

### Objective

To compare the average account balance of **high-volume** and **low-volume** transaction accounts.

### Null Hypothesis ( $H_0$ )

There is **no significant difference** in the average account balance between the two groups.

### Alternative Hypothesis ( $H_1$ )

There **is a significant difference** in the average account balance between the two groups.

### Result

- T-statistic = 1.7595
- P-value = 0.0801

### Conclusion

Since the **P-value (0.0801)** is **greater than 0.05**, the **Null Hypothesis was not rejected**.

This means there is **no statistically significant difference** in the average account balance between high-volume and low-volume transaction accounts.

```
[41]
✓ Os t_stat, p_value = ttest_ind(high_volume, low_volume, equal_var=False)

print("T-statistic:", t_stat)
print("P-value:", p_value)

... T-statistic: 1.7595094127904853
P-value: 0.08013695745786936
```

```
[42]
✓ Os if p_value < 0.05:
    print("Reject the Null Hypothesis")
else:
    print("Fail to Reject the Null Hypothesis")

Fail to Reject the Null Hypothesis
```

### Business Recommendation

- Do not evaluate customers only by transaction volume.
- Consider **account balance, risk score, account type, and customer activity** together while making business decisions

# Thank You